ANNEXURE A – AGILE SPRINT PLANNING & USER STORY WORKSHOP

PROJECT TITLE: SMART AGRICULTURE MANAGEMENT SYSTEM

PROBLEM STATEMENT

Agriculture involves several activities, including crop monitoring, irrigation scheduling, weather checking, fertilizer application, and harvesting. Managing these activities manually can lead to missed schedules, incomplete records, and delays in decision-making.

The Smart Agriculture Management System aims to provide a centralized application that helps farmers maintain crop records, plan irrigation, access weather information, manage fertilizer tasks, and record agricultural activities. The system helps organize farm operations and supports better planning through a simple, user-friendly interface.

1. IDENTIFY THE PROJECT VISION AND STAKEHOLDERS

Project Vision

To develop a centralized application that helps farmers monitor crops, organize irrigation schedules, view weather updates, manage fertilizer requirements, and track daily agricultural activities efficiently.

Project Objectives

1. Maintain crop details and monitor crop growth.
2. Create and manage irrigation schedules.
3. Display weather information for the farm location.
4. Record fertilizer tasks and application reminders.
5. Maintain agricultural activity records.
6. Provide reminders for important farming tasks.

Key Stakeholders

- **Farmers:** Manage crops, irrigation, fertilizer tasks, and activity records.
- **Agricultural Workers:** View assigned tasks and update completed activities.
- **Agricultural Advisors:** Provide guidance on crop care and fertilizer usage.
- **System Administrator:** Manage users and application settings.
- **Development Team:** Design, develop, test, and maintain the application.
- **Weather Data Provider:** Supplies weather information when an external service is integrated.

2. CREATE EPICS AND USER STORIES

An epic represents a major feature of the application. A user story describes a requirement from the user's perspective using the format: *As a user, I want a feature, so that I receive a benefit.*

Epic 1 – Crop Management

User Story: As a farmer, I want to add crop details such as crop name, planting date, and growth stage, so that I can monitor crop progress.

Epic 2 – Irrigation Scheduling

User Story: As a farmer, I want to create irrigation schedules, so that I can water crops at planned times.

Epic 3 – Weather Information

User Story: As a farmer, I want to view weather forecasts, so that I can plan agricultural activities according to expected weather conditions.

Epic 4 – Fertilizer Management

User Story: As a farmer, I want to record fertilizer tasks and reminders, so that I can complete fertilizer-related activities on time.

Epic 5 – Activity Tracking

User Story: As a farmer, I want to record sowing, watering, and harvesting activities, so that I can maintain a history of farm operations.

Epic 6 – Notifications

User Story: As a farmer, I want to receive reminders for irrigation and fertilizer tasks, so that I do not miss important activities.

3. DEFINE ACCEPTANCE CRITERIA

Acceptance criteria specify the conditions a feature must satisfy. The Given–When–Then format is used to define the expected behavior.

Crop Management

- Given the farmer is signed in, when valid crop details are submitted, then the system saves and displays the crop record.
- Given required fields are missing, when the form is submitted, then the system displays a validation message.

Irrigation Scheduling

- Given a crop is registered, when the farmer enters an irrigation date and time, then the system saves the schedule.

- Given a schedule exists, when the farmer marks it completed, then the system updates its status.

Weather Information

- Given weather data is available, when the farmer opens the weather section, then the system displays current conditions and forecast details.
- Given the weather service is unavailable, when the section is opened, then the system displays an appropriate message.

Fertilizer Management

- Given a crop is selected, when fertilizer task details and a due date are entered, then the system saves the task.
- Given a reminder is enabled, when the due time arrives, then the application displays a notification if notifications are available.

Activity Tracking

- Given activity details are entered, when the farmer saves them, then the system stores the activity record.
- Given activity records exist, when the history section is opened, then the system displays the saved activities.

4. PRIORITIZE THE PRODUCT BACKLOG

The product backlog contains all features and tasks required for the project. Features are prioritized according to user needs, importance, dependencies, and development effort.

High Priority

1. Farmer registration and sign-in.
2. Crop record creation and management.
3. Irrigation schedule creation and tracking.
4. Agricultural activity recording.

Medium Priority

1. Weather information integration.
2. Fertilizer task management and reminders.
3. Dashboard displaying upcoming tasks.

Low Priority

1. Advanced farm reports and analytics.
2. Multiple-farm comparison.

3. Soil-moisture sensor integration.
4. Automated irrigation control.

Prioritization Rationale

Crop management and irrigation scheduling are prioritized because they support the main purpose of the application. Weather information, fertilizer reminders, and advanced features can be added after the basic application is working correctly.

5. PLAN THE SPRINT BACKLOG

A sprint backlog contains the user stories and tasks selected for a development sprint.

Sprint 1 Goal

To develop a basic working application that allows farmers to access the system, add crop details, and create and view irrigation schedules.

Development Tasks

1. Design the farmer access, crop management, and irrigation screens.
2. Define the required crop information fields.
3. Implement crop record creation and viewing.
4. Implement irrigation schedule creation and display.
5. Validate required fields and handle invalid inputs.
6. Test the crop management and irrigation features.
7. Prepare a demonstration of the completed features.

Features Planned for Later Sprints

Weather integration, fertilizer management, notifications, advanced analytics, and IoT sensor integration can be developed in later sprints.

Sprint Review

At the end of the sprint, the team demonstrates the completed features and collects feedback from stakeholders. The feedback is used to improve the application and update the product backlog.

6. ESTIMATE EFFORT USING STORY POINTS

Story points estimate the relative effort, complexity, and uncertainty involved in completing a user story. The Planning Poker technique and Fibonacci sequence can be used for estimation.

Fibonacci Story Point Scale

- **1 point:** Very simple task.
- **2 points:** Small task.
- **3 points:** Moderate task.
- **5 points:** Feature requiring multiple development activities.
- **8 points:** Complex feature.
- **13 points:** Very large or uncertain feature.

Story Point Estimation

1. **Farmer Sign-In and Profile – 3 points:** Includes basic access and profile functionality.
2. **Crop Management – 5 points:** Includes crop forms, data storage, validation, and record display.
3. **Irrigation Scheduling – 5 points:** Includes date/time selection, schedule storage, and status display.
4. **Input Validation and Testing – 3 points:** Includes error handling and basic functional testing.

Total Sprint 1 Estimate:

3 + 5 + 5 + 3 = **16 story points**

These estimates are preliminary and may be adjusted after discussion among team members.

Justification

Crop management and irrigation scheduling receive 5 points each because they involve multiple operations. Farmer sign-in and testing receive 3 points each because they involve a smaller set of related tasks. Story points represent relative effort rather than fixed working hours.

7. PRESENT THE SPRINT GOAL AND EXPECTED DELIVERABLES

Sprint Goal

To provide a working prototype that enables farmers to manage basic crop information and plan irrigation through a centralized application.

Expected Deliverables

1. Basic farmer access and profile screen.
2. Crop management screen to add and view crop records.
3. Irrigation scheduling screen.

4. Irrigation task list with status updates.
5. Input validation for required fields.
6. Basic test cases and test results.
7. Sprint demonstration and stakeholder feedback.

Definition of Done

A feature is considered complete when it is implemented, tested against its acceptance criteria, integrated into the application, and demonstrated without critical errors. Any known limitations should be documented.

Expected Outcome

The first sprint is expected to deliver a working prototype with basic crop management and irrigation scheduling features. These features provide a foundation for adding weather information, fertilizer management, notifications, and other advanced functions in future sprints.

CONCLUSION

The Smart Agriculture Management System provides a centralized platform for managing crop information, irrigation schedules, weather updates, fertilizer tasks, and agricultural activities. It helps farmers organize their daily operations and make informed decisions.

Agile sprint planning enables the development team to divide the project into manageable tasks, prioritize important features, estimate effort, and deliver the application incrementally. The first sprint focuses on crop management and irrigation scheduling, while additional features can be introduced in future sprints based on stakeholder feedback and project priorities.